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THESIS TITLE

BY

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DISSERTATION

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Abstract

Write abstract here.

Write Dedication Here

Acknowledgments

Write your acknowledgements here

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List of Abbreviations

2HDM	Two-Higgs Doublet Model.
AK4	An anti-k_t jet with $R = 0.4$.
ALICE	A Large Ion Collider Experiment.
AM	Associated Memory.
AMB	Associated Memory Board.
ATCA	Advanced Telecommunications Computing Architecture.
ATLAS	A Toroidal LHC ApparatuS.
AUX	Auxiliary Board.
AXW	AUX Wrapper.
BR	Branching Ratio.
CA12	A Cambridge/Aachen jet with $R = 1.2$.
CC	Clock Correction.
CERN	European Organization for Nuclear Research.
CL	Confidence Level.
CMS	Compact Muon Solenoid.
CP	Combined Performance.
CR	Control Region.
DF	Data Formatter.
DFW	DF Wrapper.
DRAM	Dynamic Random Access Memory.
DSP	Digital Signal Processing Block.
ECAL	Electromagnetic Calorimeter.
ECR	Event Count Reset.
EGM	Extended Gauge Model.
EF	Event Filter.

ELID	Extended Level 1 Identifier.
EXP	Extrapolator.
EXTF	Extrapolator / Track Fitter.
FC	Flow Control.
FCAL	Forward Calorimeter.
FE	Front End.
FF	Flip Flop.
FIFO	First In First Out.
FLIC	FTK Level-2 Interface Crate.
FPGA	Field Programmable Gate Array.
FSR	Final State Radiation
fSSB	Final Second Stage Board.
FTK	Fast TracKer.
GTX	Gigabit Transceiver.
ggF	gluon Fusion.
HEP	High Energy Physics.
HLT	High Level Trigger.
HP	High Purity.
HVT	Heavy Vector Triplet.
HW	Hit Warrior.
HXMPP	Hit Memory++.
IBL	Insertable B Layer.
ID	Inner Detector.
IM	Input Mezzanine.
ISR	Initial State Radiation.
JVF	Jet Vertex Fraction.
JER	Jet Energy Resolution.
JES	Jet Energy Scale.
KK	Kaluza-Klein.
L1	Level 1 Hardware Trigger.
L2	Level 2 Hardware Trigger.
LAMB	Local Associated Memory Board.
LDC	Link Destination Card.

LED	Light Emitting Diode.
LFSR	Linear Feedback Shift Register.
LHC	Large Hadron Collider.
LHCb	Large Hadron Collider beauty.
LINAC2	Linear Accelerator 2.
LP	Low Purity.
LSC	Link Source Card.
LUT	Look Up Table.
MC	Monte Carlo.
MET	Missing Transverse Energy.
MIG	Memory Interface.
ML	Maximum Likelihood.
MS	Muon Spectrometer.
NLO	Next to Leading Order.
NNLO	Next to Next to Leading Order.
PDF	Parton Distribution Function.
PIX	Pixel Detector.
PU	Processing Units.
PS	Proton Synchrotron.
pSSB	Preliminary Second Stage Board.
QCD	Quantum Chromo Dynamics.
QSFP	Quad Small Form-factor Pluggable.
RAM	Read Access Memory.
RLDRAM	Reduced Latency DRAM.
ROD	Read Out Driver.
ROI	Region of Interest.
ROM	Read Only Memory.
ROS	Read Out System.
RS	Randall-Sundrum.
RTM	Rear Transmission Module.
RX	Receiver.
SCT	Semiconductor Tracker.
SE	Sync Engine.

SF	Scale Factor.
SFP	Small Form-factor Pluggable.
SLink	Super Link.
SM	Standard Model.
SPS	Super Proton Synchrotron.
SR	Signal Region.
SSB	Second Stage Board.
SSID	Super Strip Identifier.
TDAQ	Trigger and Data Acquisition.
TF	Track Fitter.
TFW	Track Fitter Wrapper.
TileCAL	Hadronic Calorimeter.
TRT	Transition Radiation Tracker.
TX	Transmitter.
VBF	Vector Boson Fusion.
VME	VERSA-Module Euro.
VR	Validation Region.

List of Symbols

ℓ	lepton.
e	electron.
μ	muon.
τ	tau.
ν_e	electron neutrino.
ν_μ	muon neutrino.
ν_τ	tau neutrino.
q	quark, also used for charge.
u	up quark.
d	down quark.
s	strange quark.
c	charm quark.
b	bottom quark.
t	top quark.
g	gluon.
γ	photon.
W	W boson.
Z	Z boson.
H	Higgs boson.
V	Vector boson.
p	proton.
n	neutron.
G^*	RS graviton.
G_{KK}	KK graviton.
W'	Heavy W boson.

Z'	Heavy Z boson.
V'	Heavy Vector Boson.
E_T^{miss}	Missing transverse energy.
E_T	Transverse energy.
p_T	Transverse momentum.
η	pseudo rapidity.
θ	polar angle.
ϕ	azimuthal angle.
d_0	the distance to the z-axis at the perigee point.
z_0	the z-coordinate at the perigee point.
ϕ_0	the azimuthal angle at the perigee point.
χ^2	statistic that measures how good the data fits the track.
R	distance in $\eta - \phi$ space.
j	small-R jet.
J	large-R jet.
g_f	fermion coupling.
g_H	Higgs and vector boson coupling.
g_ℓ	lepton coupling.
g_q	quark coupling.
σ	cross section.
m	mass.
fb	femto barn $= 10^{-43}$ m^2.
eV	electron Volt $= 1.602 \times 10^{-19}$ J.

Preface

The following is a summary of useful concepts in high energy particle physics.

0.1 Units

Discussion of units

0.2 Coordinates

LHC coordinate systems

0.3 Statistics

Statistics in particle physics

Chapter 1

Introduction

This is the first chapter of the dissertation. [1]

1.1 Creating Figures

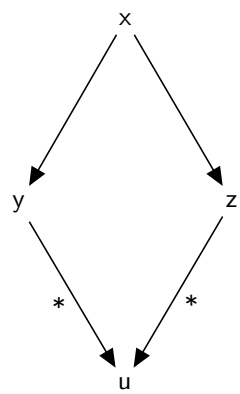


Figure 1.1: This is a placeholder figure to act as an example. Here we cite a new reference in the caption to demonstrate that given the package configuration our order of references will not be distributed by the table of contents [2].

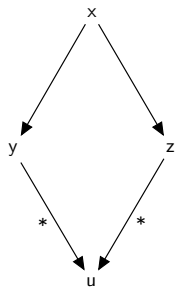
As can be seen in Figure 1.2, the subfigures are independent of each other such that Figure 1.2a and Figure 1.2b can be accessed separately.

As an example of an equation formatted in "display style" the equation for the fiducial cross section from [3] is reproduced as Equation 1.1:

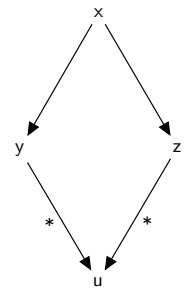
$$\sigma_{\text{inel}}^{\text{fid}}\left(\zeta > 10^{-6}\right) = \frac{N - N_{\text{BG}}}{\epsilon_{\text{trig}} \times \mathcal{L}} \times \frac{1 - f_{\zeta<10^{-6}}}{\epsilon_{\text{sel}}} \tag{1.1}$$

1.2 Creating Tables

To create tables in LaTeX it is highly recommended to use the `booktabs` package. It allows for very elegant and clean table creation, such as Table 1.1. If you want to create a table quickly, or have a CSV file that



(a) This is the first figure of two, in this example, and its own independent subfigure.



(b) As the `t` alignment option was chosen for the subfigures, they are still properly aligned vertically even though this caption is longer.

Figure 1.2: An example of a figure that consists of two subfigures.

you'd like to quickly turn into a table there are various online LaTeX table generators.

Table 1.1: Common quantities in particle physics given in both natural units and SI units.

Quantity	Natural Units	Natural Units (dimensionful)	SI Units
Speed	1	c	3.0×10^{8} m/s
Angular Momentum	1	$\hbar$	10^{34} m^2 kg/s
Energy	GeV	GeV	1.6×10^{-10} J
Momentum	GeV	GeV$/c$	1×10^{-19} kg m/s
Mass	GeV	GeV$/c^2$	1.8×10^{-27} kg
Time	1/GeV	$\hbar$/GeV	6.6^{-25} s
Length	1/GeV	$\hbar c$/GeV	2×10^{-16} m
Electric Charge	1	$e/\sqrt{4\pi\alpha_{\mathrm{em}}}$	5.3×10^{-19} C
Magnetic Field	$(\mathrm{GeV})^2$	$(\mathrm{GeV})^2/\hbar c^2$	5×10^{16} T

Good table design requires some thought and work, so it may be worth a look through some examples:

- TeX StackExchange: Tip on how to make a visually good table
- Edward Tufte endorsed example from Darkhorse Analytics

1.3 Dealing with Widows and Orphans

To reduce the difficulty of dealing with widowed text (the last line of a paragraph at the start of a page) and orphaned text (the first line of paragraph at the end of a page) the `nowidow` package is used. However, that doesn't solve the issue of orphaned section titles. The user must manually do this, but the following simple advice from TeX FAQ is recommended:

> Once you've exhausted the automatic measures, and have a final draft you want to "polish", you should proceed to manual measures. To get rid of an orphan is simple: precede the paragraph

with \`clearpage` and the paragraph can't start in the wrong place.

Appendix A

An Appendix

Appendix text goes here.

References

[1] ATLAS Collaboration, *The ATLAS Experiment at the CERN Large Hadron Collider*, *JINST* **3** (2008) S08003. 2

[2] P. W. Higgs, *Broken symmetries, massless particles and gauge fields*, *Phys. Lett.* **12** (1964) 132–133. vii, 2

[3] ATLAS Collaboration, *Measurement of the Inelastic Proton–Proton Cross Section at* $\sqrt{s} = 13$ *TeV with the ATLAS Detector at the LHC*, *Phys. Rev. Lett.* **117** (2016) 182002, [1606.02625]. 2